Insert in the circle the first initial of your family name

Department of Physics

Please Print Clearly:

Student:_____ _____

Last Name *First Name*

Student Number: _____ **Student Section:** _____

PCS 125 – Physics: Waves and Fields

Dr. A. J. Sojahrood

Midterm – July 22, 2024

Read the following attentively:

- Ensure that your name and student number are correct in the bubble sheet.S
- You will have 120 minutes to write the exam.
- The exam consists of 15 multiple-choice questions. Answer ALL questions and show all work.
- Each multiple-choice question is worth 1 mark.
- For each multiple-choice question:
  - Select only one answer choice per question on bubble sheet or it will receive a score of zero. You need to select the correct bubble or it will be marked zero score.
  - Additionally, you need to show, in the space provided in this booklet, appropriate justification for your answer. If no work or argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.
  - Transfer your choices to the bubble sheet only when you are sure of them.
  - Some answers may be rounded off.
- Pagers, cell phones, watches, and other electronic devices (other than a calculator fitting the description below) must be silenced and placed out of reach. Hats and earphones are not permitted. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.
- Sharing of pencils, erasers and calculators is not permitted.
- Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.
- Any non-program/graphing calculator models allowed for the rest are i.e. Sharp EL-546 or Casio FX-991 (or similar models). If you are found to have another type of calculator, it may be confiscated for the duration of the test.
- The exam booklet has a total of 9 pages, excluding the two-paged formula sheet. You are responsible for checking that you have all pages. You may detach the formula sheet from the booklet.
- You may not leave the exam room during the final 15 minutes of the exam.

_____

Sign here to signify you read the instructions

1. A spring-mass system is oscillating with period T. If I increase the mass four times and then stretch the spring four times as much and let it go, compared with the original period, the period of oscillation will:

a) Stays the same

b) Increase 16 times

c) Increase 2 times

d) Decrease 4 times

e) Decrease 2 times

2. A spring-mass system is in simple harmonic oscillation. Which of these statements below is correct:

a) Velocity and position always have the same direction.

b) Velocity and acceleration always have the same direction.

c) Velocity and acceleration are always in the opposite direction.

d) Acceleration and position are always in the opposite direction.

e) Velocity and position are always in the same direction.

3. We have a spring-mass oscillator. Kinetic energy is given by K. Potential Energy is given by U. At a point where the position is half the amplitude:

a) $K = U$

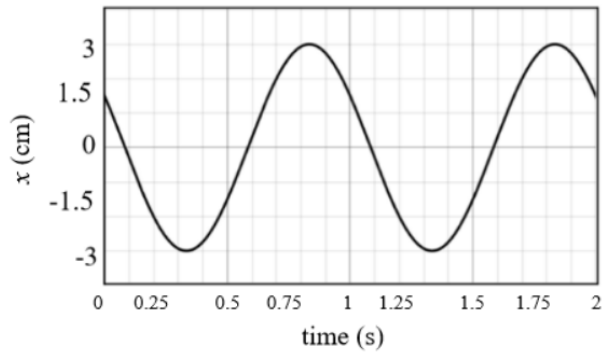
b) $K = \frac{1}{2}U$

c) $K = 3U$

d) $K = \frac{3}{4}U$

e) $K = 2U$

4. The graph below shows the oscillations as a function of time. What is the equation of the motion?



a) $x(t) = 3cm\ cos(2\pi t - \frac{\pi}{3})$

b) $x(t) = 3cm\ cos(2\pi t + \frac{\pi}{3})$

c) $x(t) = 3cm\ cos(4\pi t + \frac{\pi}{3})$

d) $x(t) = 1.5cm\ cos(2\pi t - \frac{\pi}{3})$

e) $x(t) =- 1.5cm\ cos(2\pi t - \frac{4\pi}{3})$

5. A mass-spring system is in simple harmonic oscillations with an amplitude of 6cm. If at x = -3cm, the oscillator has a velocity of 5cm/s, what is the period T of the oscillations?

a) $1\ s$

b) $2\pi\ s$

c) $0.5\ s$

d) $0.5\pi\ s$

e) There is not enough information.

6. A wave is created in the string mass system shown below. The wave number is kept constant. If we increase the mass by 4 times, the frequency of the wave:

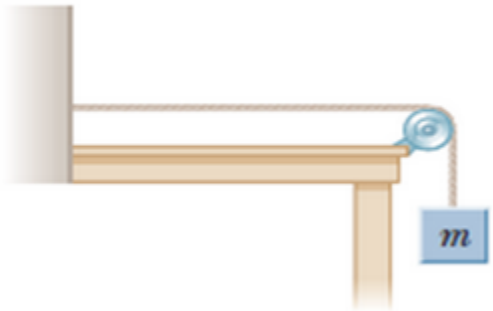
a) Increases by 4 times

b) Decreases by 4 times

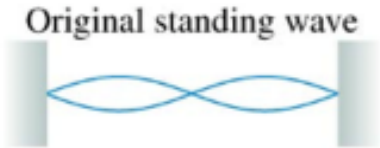
c) Increases by 2 times

d) Decreases by 2 times

e) We need more information.



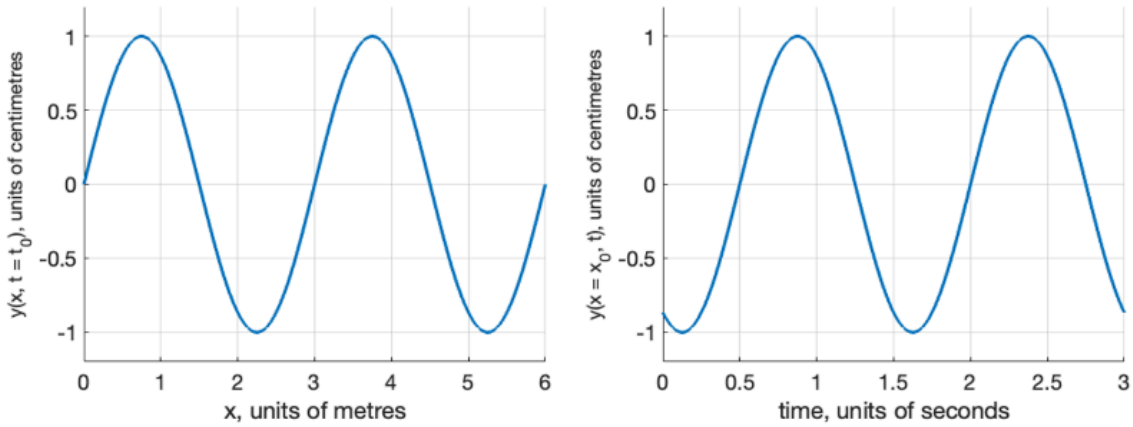
7. A standing wave on a string vibrates as shown. Suppose the string tension is reduced to $\frac{1}{16}$ its original value while the frequency and length are kept unchanged. Which standing wave pattern is produced?



Answer: D.

E) none of the above

8. The plots both show information about a single wave $y(x, t)$. The left plot shows a snapshot $y(x, t_0)$ of the wave at $t = t_0$ as the position changes. The right plot shows the wave trajectory $y(x = x_0, t)$ at position $x = x_0$ as time changes. What is the angular frequency ω of this wave?



a) 4.2 rad/s

b) 3.0 rad/s

c) 2.1 rad/s

d) 1.5 rad/s

e) 1.0 rad/s

9. Assume a change at the source of sound reduces the wavelength of a sound wave in air by a factor of 2. What happens to the speed of this sound?

a) It increases by a factor of 4

b) It increases by a factor of 2

c) It is unchanged.

d) It decreases by a factor of 2

e) It changes by an unpredictable factor.

10. An ambulance passes an observer. When it is approaching the observer, the observer hears a frequency of 500 Hz. When it moves away, the observer hears a frequency of 480 Hz. What frequency does the observer hear right at the time when the ambulance passes the observer? Assume that the speed of sound in air is 343 m/s.

a) 510 Hz

b) 500 Hz

c) 490 Hz.

d) 480 Hz

e) 470 Hz

11. I measure the sound intensity from a source to be 10 db when I am 10 meters away. What is the sound intensity when I am 1 meter away?

a) 12 dB

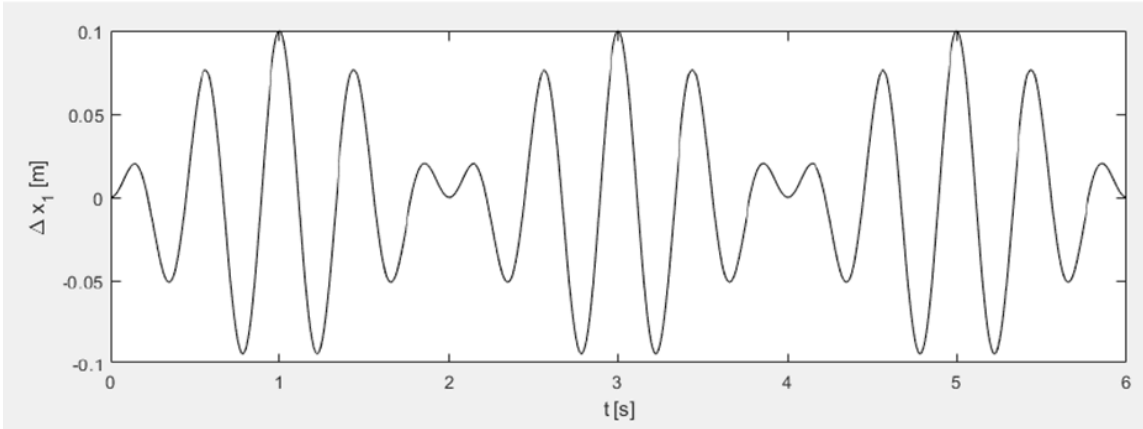
b) 30 dB

c) 8 dB

d) 20 dB

e) 0 dB

12. The waveform shown is due to the interference of two waves with different frequencies. One of the frequencies is 5.0 Hz. What is the other frequency?



a) 0.5 Hz

b) 2.5 Hz

c) 3 Hz.

d) 4 Hz

e) 4.5 Hz

13. You hear 5 beats per second when two sound sources, both at rest, play simultaneously. The beat disappears if source 2 moves away from you while source 1 remains at rest. The frequency of source 1 is 500Hz. The frequency of source 2 is:

a) 495 Hz

b) 498 Hz

c) 502 Hz.

d) 505 Hz

e) Not enough information.

14. A wave on a string is traveling to the left. At a instant, the motion of the piece of string marked with a dot is:

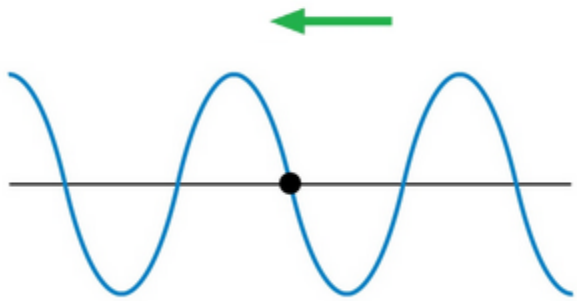
a) Up.

b) Down.

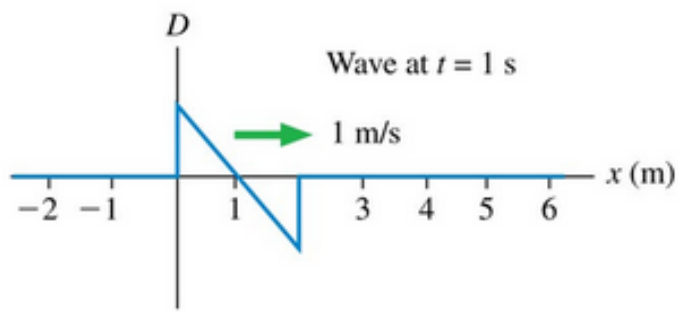
c) Right.

d) Left.

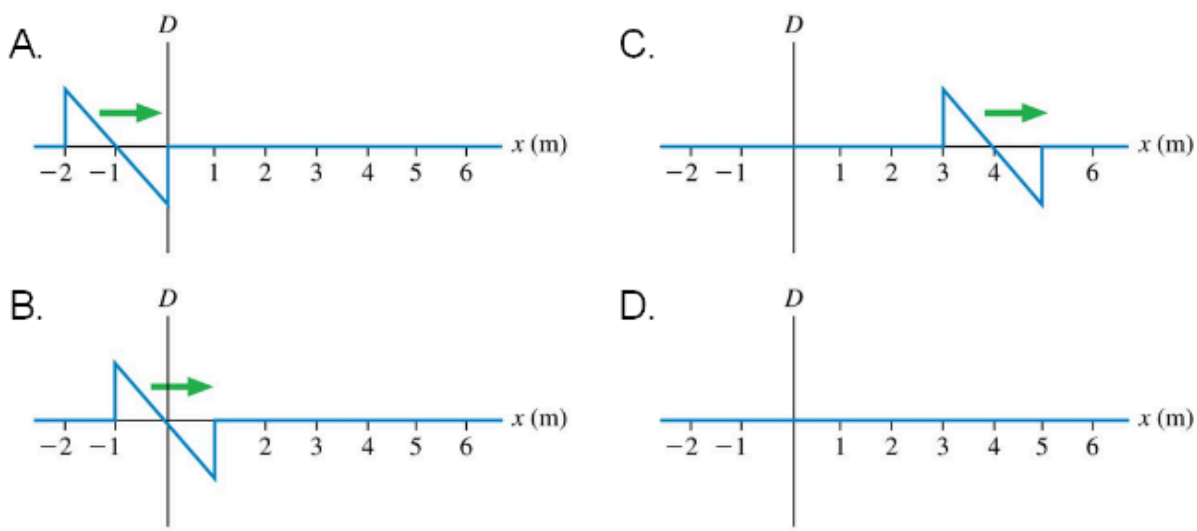
e) Zero. Instantaneously at rest.



15. This is a snapshot of a graph at $t = 1\ s$ of a wave pulse traveling to the right at 1 m/s. Which graph below shows the wave pulse at $t = 0\ s$?



Answer: b)



E) None of the above.

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PCS 125 – FORMULA SHEET

(Last Modified: February 18, 2024)

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$v = f\,\lambda$$

$$k = \frac{2\pi}{\lambda}$$

$$y_1 + y_2 = 2A\cos\left(\frac{\Delta\phi}{2}\right)\sin\left[kx - \omega t + \frac{\Delta\phi}{2}\right]$$

$$x(t) = A\cos(\omega t + \phi)$$

$$v(t) = -A\omega\,\sin(\omega t + \phi)$$

$$a(t) = -A\omega^2\cos(\omega t + \phi)$$

$$\omega = \sqrt{\frac{k}{m}};\ \omega = \sqrt{\frac{g}{L}};$$

$$\omega = \sqrt{\frac{mgd}{I}}$$

$$E = \frac{1}{2}kA^2$$

$$x(t) = Ae^{-\frac{bt}{2m}}\cos(\omega t)$$

$$\omega = \sqrt{\omega_0{}^2 - \left(\frac{b}{2m}\right)^2}$$

$$A = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0{}^2)^2 + \left(\frac{b\omega}{m}\right)^2}}$$

$$y(x,t) = A\sin[kx - \omega t + \phi]$$

$$s(x,t) = s_{max}\cos(kx - \omega t)$$

$$v = \sqrt{\frac{T}{\mu}};\ v = \sqrt{\frac{B}{\rho}};$$

$$B = -\frac{\triangle P}{\triangle V/V_i}$$

$$v = (331\ m/s)\sqrt{1 + \frac{T_C}{273°C}}$$

$$Power = \frac{1}{2}\mu\omega^2A^2v$$

$$\Delta P_{max} = \rho v\omega s_{max}$$

$$Power_{avg} = \frac{\rho A\omega^2 s^2{}_{max} v}{2}$$

$$I = \frac{\Delta P^2{}_{max}}{2\rho v}$$

$$I = \frac{Power}{A} = \frac{Power}{4\pi r^2}$$

$$y(x,t) = 2A\sin(kx)\cos(\omega t)$$

$$\Delta\phi = 2\pi\left(\frac{|r_2 - r_1|}{\lambda}\right)$$

$$\beta = (10\ dB)\,log_{10}\left(\frac{I}{I_0}\right)$$

$$f' = \left(\frac{v + v_O}{v - v_S}\right)f$$

$$f_{beat} = |f_2 - f_1|$$

$$\vec{F} = m\vec{g};\ |\vec{g}| = \frac{GM}{r^2}$$

$$F = \frac{Gm_1m_2}{r^2}$$

$$U(r) = -\frac{GMm}{r}$$

$$\vec{F} = q\vec{E}$$

$$\vec{E} = k_e\int\frac{dq}{r^2}\hat{r};\ |\vec{E}| = \frac{\sigma}{2\varepsilon_o}$$

$$F = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1q_2}{r^2}\right) = k_e\frac{q_1q_2}{r^2}$$

$$U(r) = \frac{k_eq_1q_2}{r}$$

$$\Phi_E = \oint\vec{E}\cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$$

$$U_B - U_A = q(V_B - V_A)$$

$$= -W^{Field}_{A\to B}$$

$$W^{Field}_{A\to B} = \int_A^B\vec{F}\cdot d\vec{r}$$

$$E = -\frac{dV}{dx}$$

$$V_B - V_A = -\vec{E}\cdot\vec{r}_{AB} = -El\cos\theta$$

$$= -Ed$$

$$V = \frac{k_eq}{r}$$

$$Power = I\,\Delta V$$

$$\Delta V = IR$$

$$I_{avg} = nqv_dA$$

$$R = \rho\frac{\ell}{A}$$

$$\vec{F} = q(\vec{v}\ x\ \vec{B})$$

$$\vec{F} = q\vec{E} + q\vec{v}\,x\,\vec{B}$$

$$\vec{F} = I\vec{L}\times\vec{B}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\int\frac{d\vec{s}\times\hat{r}}{r^2}$$

$$\frac{F_B}{L} = \frac{\mu_0I_1I_2}{2\pi\,a}$$

$$V_H = \frac{IB}{nqt}$$

$B = \frac{\mu_0 I}{2\,\pi\,a}$ (at a distance a from an infinitely long wire)

$B = \frac{\mu_0 I \theta}{4\,\pi\,a}$ (at the centre of an arc wire of radius a subtended by an angle θ)

$B = \frac{\mu_0\, I a^2}{2\,(a^2 + x^2)^{3/2}}$ (at a distance x along the centre line axis of a circular loop of wire radius a)

Constants

Grav. accel. on Earth	$g = 9.81\ \mathrm{m \cdot s^{-2}}$
Speed of light in vacuum	$c = 3.00 \times 10^{8}\ \mathrm{m \cdot s^{-1}}$
Gravitational constant	$G = 6.6742 \times 10^{-11}\ \mathrm{N \cdot m^{2} \cdot kg^{-2}}$
Coulomb constant	$k_e = 8.9876 \times 10^{9}\ \mathrm{N \cdot m^{2} \cdot C^{-2}}$
Elementary charge	$e = 1.6022 \times 10^{-19}\ \mathrm{C}$
Electron volt	$\mathrm{eV} = 1.6022 \times 10^{-19}\ \mathrm{J}$
Mass of electron	$m_e = 9.1094 \times 10^{-31}\ \mathrm{kg}$
Mass of proton	$m_p = 1.6726 \times 10^{-27}\ \mathrm{kg}$
Permeability of vacuum	$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m \cdot A^{-1}}$
Permittivity of vacuum	$\varepsilon_0 = 8.8542 \times 10^{-12}\ \mathrm{C^{2} \cdot N^{-1} \cdot m^{-2}}$
Density of air	$\rho_{\mathrm{air}} = 1.2\ \mathrm{kg \cdot m^{-3}}$
Threshold of hearing	$I_0 = 1 \times 10^{-12}\ \mathrm{W \cdot m^{-2}}$

Math Identities

$$sin(a \pm b) = sin\ a\ cos\ b\ \pm\ cos\ a \sin b$$

$$sin\ a \pm sin\ b = 2\cos\left(\frac{a \mp b}{2}\right)\sin\left(\frac{a \pm b}{2}\right)$$